

August 4, 2026 – PCB Integration and Troubleshooting Update

Over the past several days, I have been assembling, wiring, and testing my custom telemetry PCB. Before installing it on the RC crawler, I tested the board off the vehicle and confirmed that the system was functioning. However, after installing it on the crawler, the ESP32 began overheating, all of the sensors disconnected, and the board eventually stopped communicating completely. Once the ESP32 overheated, I could no longer upload code or establish a connection, forcing me to purchase a replacement.

After installing a new ESP32, the same problem occurred. I began systematically troubleshooting the hardware using a multimeter. I discovered that the buck converter output voltage was set too high, so I adjusted it to a regulated 5.0 V. I also found that the damaged ESP32 had developed a short between its 3.3 V rail and ground by measuring continuity and resistance between the two pins.

Additionally, I temporarily removed the battery-voltage sensing circuit because it appeared to be contributing to the ESP32 failures. With the voltage sensing removed, I was able to successfully restore operation of the Hall-effect sensor, DS18B20 temperature sensor, BNO055 IMU, Bluetooth communication, and the mobile telemetry dashboard.

While troubleshooting, I also wanted to verify that my custom PCB was not the source of the failures. I powered the PCB with the buck converter and used a multimeter to measure the voltage at the power pins throughout the board. This allowed me to confirm that the correct voltages were present at the PCB before installing another ESP32.

Finally, I completed an initial lifted-wheel test of the crawler. The telemetry system successfully calculated an unloaded theoretical maximum speed of **17.28 mph**. A ground-speed test is still required to determine the vehicle's actual speed under load.

